BRAINSTORM GROUP (BSG) MAT111 TEST (2021/22)

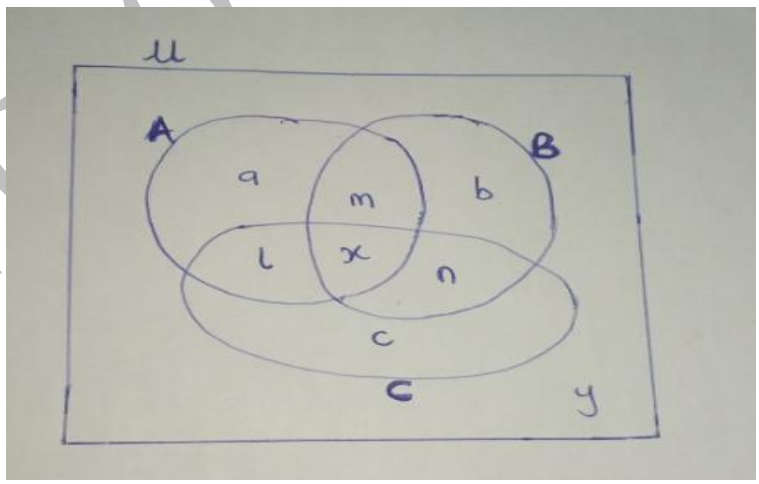
Answer all questions

1. In an examination, 18 students passed MAT111, 17 students passed PHY113, 11 students passed both subject. Find the number of students that passed Mat111.
   A .6 B. 7 C. 8 D. 9
2. If $X = \{2,\ 4,\ 6\}$, determine the power set of X, denoted by $n(P(x))$.
   A. 3 B. 6 C. 8 D. 9
3. Given that $X = \{x : 3 < x < 6\}\ and\ Y = \{x : 4 \leq x \leq 8\}.\ Find\ X \cap Y.$
   A. {4} B. {4, 5}
   C. {5, 6} D. {4, 5, 6, 7, 8}
4. If Q = {all perfect squares less than 30} and P = {all odd numbers from 1 to 10}. Find $(Q \cap P)$.
   A. {0, 1, 4, 9} B. {1, 4, 9}
   C. {1, 4} D. {4, 9} E. {1, 9}
5. If set $F = \{2 < x \leq 5\} \cup \{9 < x < 12\}$. List the members of F.
   A. $\emptyset$ B. {2, 3, 4, 5, 9, 10, 11, 12}
   C. {3, 4, 5, 6, 9, 10, 11} D. {3, 4, 5, 10, 11}
6. Given that $P = \{1,\ 3,\ 5,\ 7\}\ and\ Q = \{2,\ 5,\ 6,\ 8,\ 9\}.\ Find\ (P \cup Q).$
   A. {1, 2, 3, 5, 6, 7, 8, 9} B. {1, 2, 3, 6, 7, 8, 9}
   C. {1, 2, 3, 4, 5, 6, 7, 8, 9} D. {3, 5}
7. Given that $P = \{1,\ 3,\ 5,\ 7\}\ and\ Q = \{2,\ 5,\ 6,\ 8,\ 9\}.\ Find\ (P \cap Q).$
   A. {} B. {3, 5} C. {5, 7}
   D. {3, 5, 7} E. {5}
8. If $U = \{1,\ 2,\ 3,\ 6,\ 9, 18\},\ X = \{2,\ 6,\ 18\}, Y = \{1,\ 3,\ 6\}, list\ the\ elements\ of\ X' \cap Y.$
   A. {1, 3} B. {1, 3, 6}
   C. {1, 3, 9} D. {1, 3, 6, 9}
9. In a class of 40 students, 30 take Agriculture and 20 take Physics. If 8 students take neither Agriculture nor Physics, find the number of students who take Agriculture but not Physics. A. 2 B. 8 C. 12 D. 18
10. Two sets A and B are such that n(A) = 11 and n(B) = 6. Given that $n(\mathcal{U}) = 15$, find the smallest possible value of $n(A \cap B)$.
    A. 0 B. 17 C. 5 D. 2
11. Two sets A and B are such that $n(A) = 11$ and $n(B) = 6$. Given that $n(\mathcal{U}) = 15$, find the largest possible value of $n(A \cup B)'$
    A. 2 B. 5 C. 6 D. 4
12. All inhabitants of a certain village near the Cameroon border speak either French or English or both. If 64% speak French and 58% speak English. What is percentage speak both languages?
    A. 18 B. 22 C. 23 D. 25
13. The $A \cup B$ is 69, the number of A is 40 and the number of B is 54. How many people are in $A \cap B$?
    A. 15 B. 25 C. 27 D. 29
14. In a class of 50 students, 25 like oranges, 22 like bananas and 4 like both oranges and bananas. How many like neither oranges nor banana?
    A. 1 B. 7 C. 11 D. 12
15. In a class, 16 students offer geography, 18 offer chemistry, 5 offer both and 3 offer neither. How many students are in the class?
    A. 27 B. 33 C. 32 D. 42
16. Given that U = $\{x : x \text{ is an integer}, 2 \leq x \leq 10\}, A = \{x : x\ is\ a\ prime\ number\},\ B = \{x:\ is\ a\ multiple\ of\ 3\}. Find\ (A \cup B).$
    A. {3} B. {2, 3, 5}
    C. {3, 5, 6, 7} D. {2, 3, 5, 6, 7, 9}
17. Given that U = $\{x : x \text{ is an integer}, 2 \leq x \leq 10\}, A = \{x : x\ is\ a\ prime\ number\},\ B = \{x:\ is\ a\ multiple\ of\ 3\}. Find\ n(A \cup B).$
    A. 1 B. 3 C. 4 D. 6
18. Given that U = $\{x : x \text{ is an integer}, 2 \leq x \leq 10\}, A = \{x : x\ is\ a\ prime\ number\},\ B = \{x:\ is\ a\ multiple\ of\ 3\}.$ List the elements of $A' \cap B'$.
    A. {2, 3, 5, 6} B. {4, 8, 9, 10}
    C. {2, 3, 5, 6, 7, 9} D. {4, 8, 10}
19. In a class of 60 students, 40 speak Hausa while 19 speak Igbo and 7 students speak both languages. How many student speak neither of the languages.
    A. 1 B. 8 C. 15 D. 16
20. P and Q are two sets such that is n$(P \cup Q)$ is 50, n(P) is 40 and n(Q) is 32. What is n$(P \cap Q)$?
    A. 10 B. 12 C. 18 D. 22
21. Which of the following is not a subset of P if P = {2, 4, 6}.

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A. {2, 2, 4} B. {2, 3, 4}
C. {2, 4, 6} D. ∅

22. $(M' \cap N)'$ is equivalent to
A. $(M \cap N')'$ B. $(M' \cap N)$
C. $(M \cup N')$ D. $(M \cup N)$
23. If $P \subset Q, then\ (P \cup Q) \cap P\ is$ is
A. ∅ B. P C. Q D. $\mathcal{U}$
24. The cardinality of the set Z = {M, I, S, S, I, S, S, I, P, P, I} is
A. 4 B. 6 C. 8 D 11
25. $(A \cap A')'$ is equivalent to
A. ∅ B. A C. A' D. $\mathcal{U}$
26. $(A \cup A')$ is equivalent to
A. ∅ B. A C. A' D. $\mathcal{U}$
27. If X = {1, 2, 3, 4} and Y = {3, 5, 6}, the elements of $(X \cap Y) \cup$ X are
A. {3} B. {1, 2, 3, 4}
C. {1, 2, 5, 6} D. {1, 2, 3, 4, 5, 6}
28. If X = {a, e, m, p}, Y = {a, e, i, o, u}, Z = {l, m, o, p, q, r, s, t, u}. The elements of $X \cup (Y \cap Z)$ are
A. {a, e, m, p, o} B. {a, e, m, p, u}
C. {a, e, m, p, o, u, t} D. {a, e, m, p, o, u}
29. If $\mathcal{U}$ = {5, 6, 7, 8}, P = {6, 7} and Q = {6, 8} then $(P \cap Q)'$ is
A. {6} B. {7, 8}
C. {6, 7, 8} D. {5, 7, 8}
30. $P - Q$ is same as
A. $(P \cap Q)$ B. $(P \cap Q)'$
C. $(P \cap Q')$ D. $(P' \cap Q)$
31. Given $n(A - B) = 18, n(A \cup B) = 70\ and\ n(A \cap B) = 25. Find\ n(B)$.
A. 27 B. 43 C. 45 D. 52*
32. In a class of 500 students, 270 offer Chemistry, 250 offer Geography, 110 offer neither. Find the number of students who offer Chemistry and Geography.
A. 110 B. 120
C. 130* D. 140
33. Sets A and B are defined by A = {3, 5, 7, 8} and B = {x, y, z}. Is x ∈ A?
A. True B. False*
34. In a Survey of 23 young man, it was recorded that 6 do play Football, 8 play Basketball and 12 play none. Find the number of young men who play only Basketball.
A. 3 B. 5 C. 7 D. 11
35. In a class of 40 students, 32 offer Mathematics, 24 offer Biology and 4 offer neither Biology nor Mathematics. Find the number of students who offer only one subject.
A. 4 B. 12 C. 16* D. 20
36. Given the set {R = x : x ∈ ℕ, x < 5}. List elements of R.
A. {… -1, 0, 1, 2, 3, 4, 5} B. {0, 1, 2, 3, 4, 5}
C. {… -1, 0, 1, 2, 3, 4} D. {1, 2, 3, 4}
37. In a class of 50 students, 31 offer Chemistry, 27 offer Biology while 3 neither. Find the number of students that offer chemistry only.
A. 11 B. 14 C. 16 D. 20



The Venn diagram above represents a group of students offering course A, B and C. Use the diagram above to answer question 38 - 40

38. The portion y can be represented as
A. (A ∪ B ∪ C) B. (A ∩ B ∩ C)
C. (A ∪ B ∪ C)' D. (A ∩ B ∩ C)'
39. Which of the following corresponds to number of students that offer only two courses?
A. (a + b + c) B. (l + m + n)
C. (x) D. (y)
40. The number of students that offer at least two courses is
A. (a + b + c + l + m + n) B. (l + m + n + x)
C. (a + b + c + x) D. (x – y)

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ANSWER							
1	B	11	D	21	B	31	D
2	C	12	B	22	C	32	C
3	B	13	B	23	B	33	B
4	E	14	B	24	A	34	B
5	D	15	C	25	D	35	C
6	A	16	D	26	D	36	D
7	E	17	D	27	B	37	D
8	A	18	D	28	D	38	C
9	C	19	B	29	D	39	B
10	D	20	D	30	C	40	B

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